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Agriculture: Rice Pest Management Guidelines

Rice Blast

Rice blast: *Magnaporthe oryzae* (anamorph=*Pyricularia oryzae*), *Pyricularia grisea*

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Symptoms

The rice blast fungus may infect and produce lesions on any aboveground rice plant tissues. The common name given to the disease depends on the tissues affected. For example, the disease is called [leaf blast](#) when lesions are observed on leaves or neck blast when lesions occur on the [panicle](#) node.

Leaf Symptoms

Lesions that occur on the leaf are usually [diamond shaped](#) with a gray or white center and brown or reddish-brown border and are 0.4 to 0.6 inch (10–15 mm) long and 0.12 to 0.2 inch (3–5 mm) wide. [Newly formed lesions](#) may have a white or gray-green center and a darker-green border. Their shape, color, and size can vary depending on varietal resistance, age of the plant, and lesion age. Lesions on a leaf can coalesce and cover large areas of the leaf tissue. Leaf blast may sometimes kill young plants up to the tillering stage. Leaf blast usually increases early in the season, then declines late in the season as leaves mature and become less susceptible. Because excess nitrogen can increase plant susceptibility to blast, leaf blast is often first observed on field headlands, where the fertilizer equipment turns, overlapping nitrogen fertilizer. Leaf blast in these areas may cause large areas of dead plants on susceptible varieties.

Leaf Collar Symptoms

Infection at the junction of the leaf blade and the [sheath](#) results in the typical brown [collar rot](#) symptom. Severe collar infection often results in the death of the [leaf](#).

Node Symptoms

Stem nodes may be infected as the plant approaches maturity, causing the complete death of the entire stem above the infection. Diseased nodes are [brown or black](#).

Panicle and Grain Symptoms

Infections just [below the panicle](#), usually at the [neck node](#), cause a neck rot or rotten neck symptom that can be very injurious to the crop and is typically referred to as neck blast. If neck blast occurs early, the [entire panicle](#) may die prematurely, leaving it white and completely blank with unfilled kernels. [Infections that occur later in the season](#) may cause incomplete grain filling and poor milling quality. Other parts of the panicle, including panicle branches and [glumes](#), may also be infected. Panicle lesions are usually brown but may also be black.

Comments on the Disease

Rice blast was first identified on California rice in 1996. The disease is favored by long periods of free moisture, high humidity (greater than 90%), little or no wind at night, and night temperatures above 68°F. Leaf wetness from dew or other sources is required for infection. The optimum temperature for spore germination, infection, lesion formation, and [sporulation](#) is 77° to 82°F. Sporulation is greatest when relative humidity is above 93%.

Shortly after infecting the plant, the fungus produces spores called [conidia](#). These spores disperse in the air and, under favorable conditions, may cause new infections on susceptible host tissue when there is a sufficient period of free moisture and favorable temperatures for completion of the infection process. When conditions are favorable, a single disease cycle can be completed in about a week. In addition, a single lesion can produce thousands of spores in one night and may

The blast fungus can overwinter from one season to the next on infected seed and diseased crop residue. Infected seed is the most important source of the fungus, followed by inoculum from overwintered and infected straw or stubble.

Management

Rice blast management in California requires implementing a variety of cultural practices: destroying infested crop residue, using pathogen-free seed, planting resistant or tolerant varieties, water seeding, continuous flooding, and avoiding excessive nitrogen fertilization. Environmental conditions in California appear to be permissive for blast development but are usually not conducive. Fungicides are available but are only needed when unusually wet conditions occur during the California growing season.

Cultural Control

Excessive nitrogen fertilization, aerobic conditions, and drought stress are all factors that can increase the risk of rice developing blast. High nitrogen rates, especially in the form of nitrate, cause rice plants to produce rapid new tissue growth that is more susceptible to the pathogen than older tissue. Extended drain periods during the season may also encourage the disease by aerating the soil which increases the conversion rate of ammonium to nitrate, and by causing drought stress to rice.

Plant [certified seed](#) to ensure that seed originates from seed fields inspected for the disease. Water seeding and maintaining a continuous flood helps reduce potential disease transmission from infected seed to seedlings. Drill seeding has a higher risk of blast disease development because of the way water is managed in this production system.

Continuous flooding limits blast development by reducing the availability of nitrate in the soil and by reducing drought stress on the plants. Avoid field drainage, especially for extended periods. Studies in other areas demonstrate that shallow water is more favorable to blast development than deeper water and blast may be more severe in shallower parts of a field. Laser leveling of fields results in more precise water management and allows for uniform water depths throughout a field, reducing the potential for high spots that may prove more conducive to disease development.

Varietal Susceptibility

Variety M-210 is resistant to the current races of the rice blast pathogen in California. All other California varieties are susceptible. Varieties M-105 and M-209 are considered very susceptible and are not recommended in areas where blast may be severe, mostly in the northern part of the Sacramento Valley. Similarly, use the resistant variety M-210 if dry or drill seeding.

Organically Acceptable Methods

All cultural controls discussed above are acceptable in a certified organic crop.

Monitoring and Treatment Thresholds

In-season monitoring determines the need for fungicide treatment. Generally, a treatment for leaf blast is not necessary unless plants are being burned down in large areas of the field. However, if leaf blast is observed, a treatment during the early heading stage may be appropriate to protect panicles from neck blast. Throughout the season, examine plants in several locations throughout the field for the presence of leaf lesions and intensify monitoring as plants approach the [boot](#) stage.

Because blast spores can travel long distances, fields that show no symptoms of leaf blast may still develop neck blast during heading later in the season. If blast is present in the area, a treatment at the early heading stage may be justified to protect panicles.

When making a treatment decision, consider disease progress, crop growth stage, environmental conditions, and rice variety. For example, there is a greater risk of neck and panicle blast infections occurring when growing one of the more susceptible varieties, actively sporulating leaf lesions are present, long periods of leaf wetness occur each day, and warm night temperatures persist over several days. Use a fungicide to protect panicles as they emerge from the boot.

Common name	Amount per acre	REI‡	PHI‡
(Example trade name)		(hours)	(days)

Pesticide precautions

Protect water

Calculate VOCs

Protect bees

choosing a pesticide, consider information relating to the pesticide's properties and application timing, [honey bees](#), and [environmental impact](#). Always read the label of the product being used.

A. AZOXYSTROBIN

(Quadris)	12.5–15.5 fl oz	4	28
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MODE-OF-ACTION GROUP NAME (NUMBER¹): Quinone outside inhibitor (11)

COMMENTS: Limited studies have shown a single application at the lowest labeled rate to be effective, but results may vary under different conditions. Can be applied as a preventive treatment for blast control and applied before favorable conditions for blast development. For panicle blast, application can be made at mid-boot to boot-split, but before full head emergence. Under heavy disease pressure and conditions favorable for disease development, use maximum label rates; a second application may be applied when panicles are about 60 to 90% emerged from the boot (7–14 days later). Water holding period is 14 days.

‡ Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without protective clothing. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases, the REI exceeds the PHI. The longer of the two intervals is the minimum time that must elapse before harvest.

¹ Group numbers are assigned by the [Fungicide Resistance Action Committee \(FRAC\)](#) according to different modes of action. Fungicides with different group numbers are suitable to alternate in a resistance management program. In California, make no more than one application of a fungicide with a mode-of-action group number associated with high resistance risk before rotating to a fungicide with a different mode-of-action group number; for other fungicides, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action group number.



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